## Exercise 1: Cruise

A turbojet aircraft shall fly a distance of s = 2000 km with a velocity of V = 65 m/s with the least amount of fuel. Throughout the flight the lift coefficient is constant! The following data is given:

| | | |
|---|---|---|
| Aircraft mass (including fuel) at the beginning of the flight | $m$ | $2000\ \mathrm{kg}$ |
| k-factor | $k$ | $0{,}018$ |
| Reference area | $S_{ref}$ | $18\ \mathrm{m}^2$ |
| Zero-lift drag coefficient | $C_{D0}$ | $0{,}01$ |
| Specific fuel consumption w.r.t. engine performance parameter | $b_L$ | $7{,}6 \cdot 10^{-6}\frac{kg}{N}$ |

What is the initial and final height? How long does the flight take?

Hint: Assume a symmetric quadratic drag polar.

## Exercise 2: Equations of Motion

A turbojet aircraft shows the following characteristics:

| | | |
|---|---|---|
| Maximum engine performance parameter | $L_{Pmax}(\rho)$ | $5000\ \mathrm{N}$ |
| Zero-lift lift coefficient | $C_{L0}$ | $0{,}01$ |
| Lift derivative | $C_{L\alpha}$ | $6{,}2$ |
| Aircraft mass | $m$ | $2000\ \mathrm{kg}$ |
| k-factor | $k$ | $0{,}02$ |
| Reference area | $S_{ref}$ | $18\ \mathrm{m}^2$ |
| Zero-lift drag coefficient | $C_{D0}$ | $0{,}02$ |
| Maximum lift coefficient | $C_{Lmax}$ | $1{,}2$ |

For a true airspeed of $V_{TAS} = 60m/s$ the pilot would like to start a steady-state climb (straight and without sideslip) with $\gamma_K = 3°$. For this he puts the thrust lever to $\delta_T = 0.5$.

What is the resulting angle of attack? Use the rigid-body equations of motion to state the result.
Hint: You can assume the density $\rho$ to be constant.

Additional data:

| | |
|---|---|
| Density at MSL according to ISA standard atmosphere: | $\rho_{s,MSL} = 1{,}225 \frac{kg}{m^3}$ |
| Pressure at MSL according to ISA standard atmosphere: | $p_{s,MSL} = 1{,}01325e5\ Pa$ |
| Temperature at MSL according to ISA standard atmosphere: | $T_{s,MSL} = 288{,}15\ K$ |
| Gravitational acceleration: | $g_0 = 9{,}80665 \frac{m}{s^2}$ |
| Specific gas constant: | $R = 287{,}05 \frac{J}{kg\ K}$ |
| Polytropic exponent: | $n = 1{,}235$ |
| Isentropic exponent: | $\kappa = 1{,}4$ |
| Radius of the earth: | $r_{earth} = 6356766\ \mathrm{m}$ |